INCH-POUND

MIL-S-20708/49C
13 March 1989

SUPERSEDING
MIL-S-20708/49B
30 July 1987

MILITARY SPECIFICATION SHEET

SYNCHRO, DIFFERENTIAL RECEIVER, TYPE 23TDR4B

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Synchros described herein shall consist of this specification and the latest issue of MIL-S-20708.

TABLE I. Requirements.

Requirement	Value	Unit	Tolerance
Frequency	400	Hz	±1%
Primary Voltage	78 Ⓒ	volts	±1%
Primary Current	950.0	milliamps	maximum
Primary Power	5.2	watts	maximum
Impedance			
Zp	----	ohms	min.-max.
Zd	----	ohms	min.-max.
Impedance Angle:			
Zp	----	degrees	min.-max.
Zd	----	degrees	min.-max.
Transformation Ratio	1.154	------	±2%
Phase Shift (Lead)	4.0	degrees	maximum
Electrical Error	6.0	minutes	maximum
Receiver Error	45.0	minutes	maximum
Synchronizing Time: 30° ±2°	1.0	seconds	maximum
Synchronizing Time: 177° ±2°	2.0	seconds	maximum
Torque Gradient	0.16	ounce-inches per degree	minimum
Radial Play	0.0010	inches	maximum
End Play	0.0050	inches	maximum
Temperature Rise	25.0	degrees (C)	maximum
Variation of Voltage (+10%) and Freq. (-5%)	----	watts	maximum

Ⓒ denotes changes.

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TABLE II. Military part number variant characteristics.

Military Part No. 1/	A ±.010	B ±.010	G Maximum	Shaft Style	Terminal End	U +.0000 −.0002
M20708/49-01B	.676	.719	3.410	Splined	Terminal Board	.2405

1/Military part number M20708/49-01B shall be the NATO standard.

1/The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation.

TABLE III. Military part number cross-references.

SUPERSEDED MILITARY PART NUMBERS		NEW MILITARY PART NUMBERS
MIL-S-20708/49A	MIL-S-20708/49	
M20708/49A-001	M20708/49-001	M20708/49-01B

NOTE: All line items shown in Table III refer to equivalent and interchangeable synchros of the same type designation modification. Part number changes do not affect form, fit, or function of the synchros listed therein.

Custodians:
Navy-AS

Review Activities:
Air Force-99
DLA-ES

User Activities:

Preparing Activity:
Navy-AS

(Project 5990-0349-32)

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